

SUTHERLAND, South Africa

WMO: 687220

Lat: 32.400S

Lon: 20.663E

Elev: 1458

StdP: 84.99

Time Zone: 2.00 (E02)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-5.9	-4.3	-13.5	1.4	14.4	-11.7	1.6	12.4	9.8	9.3	9.0	12.1	0.5	210	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.8	31.6	14.1	30.1	13.6	28.8	13.2	16.7	23.9	16.1	23.7	15.4	23.5	3.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	12.5	18.9	13.7	11.7	18.8	12.5	10.8	18.4	52.8	24.0	50.6	23.6	48.4	23.4	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.4	6.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	20.3	19.7	17.4	13.4	9.8	6.4	6.0	6.7	9.9	13.0	15.4	18.4		
	DBStd	6.01	2.81	2.98	3.25	3.55	3.11	2.96	3.57	3.10	3.09	3.95	3.75	3.04		
	HDD10.0	467	0	0	1	10	42	115	131	111	38	14	4	1		
	HDD18.3	2162	13	18	55	151	264	358	381	361	252	171	101	36		
	CDD10.0	1567	319	272	232	112	36	7	9	9	37	108	166	262		
	CDD18.3	220	74	57	27	3	0	0	0	0	6	13	39			
	CDH23.3	3131	911	719	368	63	6	0	0	1	17	164	277	604		
	CDH26.7	893	309	221	82	4	0	0	0	1	41	62	174			
(14) Wind	WSAvg	2.7	2.8	2.6	2.5	2.4	2.6	3.0	2.8	2.7	2.7	2.9	2.9	2.8		
(15) Precipitation	PrecAvg	263	14	18	32	28	24	36	26	24	17	16	18	15		
	PrecMax	495	115	93	111	103	87	110	86	64	79	63	89	86		
	PrecMin	132	0	0	0	1	1	0	2	0	0	0	0	0		
	PrecStd	78	21	20	28	27	24	27	21	17	16	16	21	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	33.1	31.1	27.1	24.0	18.7	19.8	22.9	25.6	31.2	31.0	32.2		
		MCWB	14.7	14.9	14.3	12.8	10.8	8.1	8.3	9.5	10.3	13.2	12.8	14.0		
	2%	DB	31.6	30.9	29.0	25.1	21.0	16.7	17.1	18.6	22.8	27.3	28.2	30.2		
		MCWB	14.2	14.2	13.4	12.0	9.7	7.7	7.8	7.8	9.6	11.6	12.1	13.2		
	5%	DB	29.9	29.4	27.3	23.1	18.4	14.5	15.3	16.2	20.4	25.1	26.4	28.7		
		MCWB	13.5	13.8	13.0	11.6	8.8	6.8	6.7	6.8	8.2	10.8	11.6	12.8		
	10%	DB	28.3	27.8	25.5	20.8	16.3	12.4	13.3	13.9	18.0	22.7	24.6	26.9		
		MCWB	13.4	13.4	12.6	10.9	8.4	6.1	6.3	6.0	7.6	10.0	11.0	12.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	17.4	18.0	15.2	12.4	10.5	9.7	10.6	11.7	14.7	15.6	16.5		
		MCDB	23.4	24.9	26.2	22.4	16.8	15.0	16.5	16.9	23.0	23.3	24.3	23.5		
	2%	WB	16.6	16.5	16.6	14.1	11.4	8.9	8.5	9.1	10.2	13.6	13.7	15.6		
		MCDB	23.7	24.8	22.7	20.8	16.5	13.0	15.0	17.5	20.0	23.9	23.5	23.2		
	5%	WB	15.9	15.8	15.4	13.3	10.5	7.9	7.6	7.6	9.1	12.3	12.8	14.8		
		MCDB	24.1	24.2	22.0	19.9	15.7	12.6	13.8	14.4	17.8	21.5	23.1	23.4		
	10%	WB	15.3	15.1	14.3	12.1	9.6	7.0	6.7	6.4	8.1	11.1	11.8	14.1		
		MCDB	24.0	23.7	22.2	18.6	14.8	11.6	12.1	13.0	16.5	20.4	22.1	23.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	15.4	15.4	14.3	12.8	11.4	12.4	13.7	15.4	15.9	15.3	15.3		
		MCDBR	17.5	17.7	17.8	18.0	17.0	15.5	15.9	17.8	20.0	20.5	18.8	18.4		
		MCWBR	5.2	6.1	6.4	7.5	8.2	8.4	8.3	8.6	9.1	8.2	6.4	6.2		
		MCDBR	13.2	14.4	13.6	13.8	12.6	12.3	13.8	15.2	17.3	16.8	15.9	13.7		
	5% WB	MCWBR	4.6	5.2	5.0	6.2	6.8	7.3	7.8	8.1	8.4	6.8	5.9	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.288	0.283	0.268	0.255	0.244	0.247	0.266	0.278	0.292	0.277	0.281		
	taud		2.635	2.629	2.639	2.700	2.723	2.719	2.693	2.592	2.520	2.509	2.628	2.640		
	Ebn at Noon		1055	1033	1006	974	941	930	941	959	994	1017	1057	1062		
	Edh at Noon		100	97	90	75	65	61	66	82	99	108	100	100		
(44) All-Sky Solar Radiation	RadAvg		8.55	7.70	6.34	4.75	3.55	3.01	3.35	4.22	5.72	7.13	8.22	8.82		
	RadStd		0.25	0.25	0.20	0.24	0.19	0.19	0.16	0.19	0.24	0.26	0.29	0.36		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	-1.04	N/A	N/A	N/A	N/A

Nomenclature: See separate page